WaveWash

User Manual

Check, repair and clean your audio files before you train on them.

Version 0.1.0a1 (alpha). Free and open source (MIT).
Written in plain English, step by step.

What it does, in 3 lines

• You point it at a folder of audio files and say what you are training (speech-to-text, a voice, speaker ID, music).
• It checks every file. Good files go to one folder, bad files go to another with the reason written on them.
• It sets duplicates aside and writes a report for each file. **Your original files are never changed or deleted.**

Contents

• 1. Install (one time)
• 2. Start the app
• 3. Use the screen, step by step
• 4. Read your results
• 5. Using the command line instead
• 6. Common problems
• 7. What WaveWash can and cannot do

1. Install (one time)

You need Python 3.10 or newer. Open Terminal (Mac) or Command Prompt (Windows) and type:

```
pip install git+https://github.com/BoobalanArjunan/wavewash.git
```

Once WaveWash is on PyPI, you will be able to use `pip install --pre wavewash` instead. The `--pre` is needed because this is an alpha version.
Check that everything is fine:

```
wavewash doctor
```

It lists what is installed. The last line should say **Result: all good**. You do not need Node or any web tools.

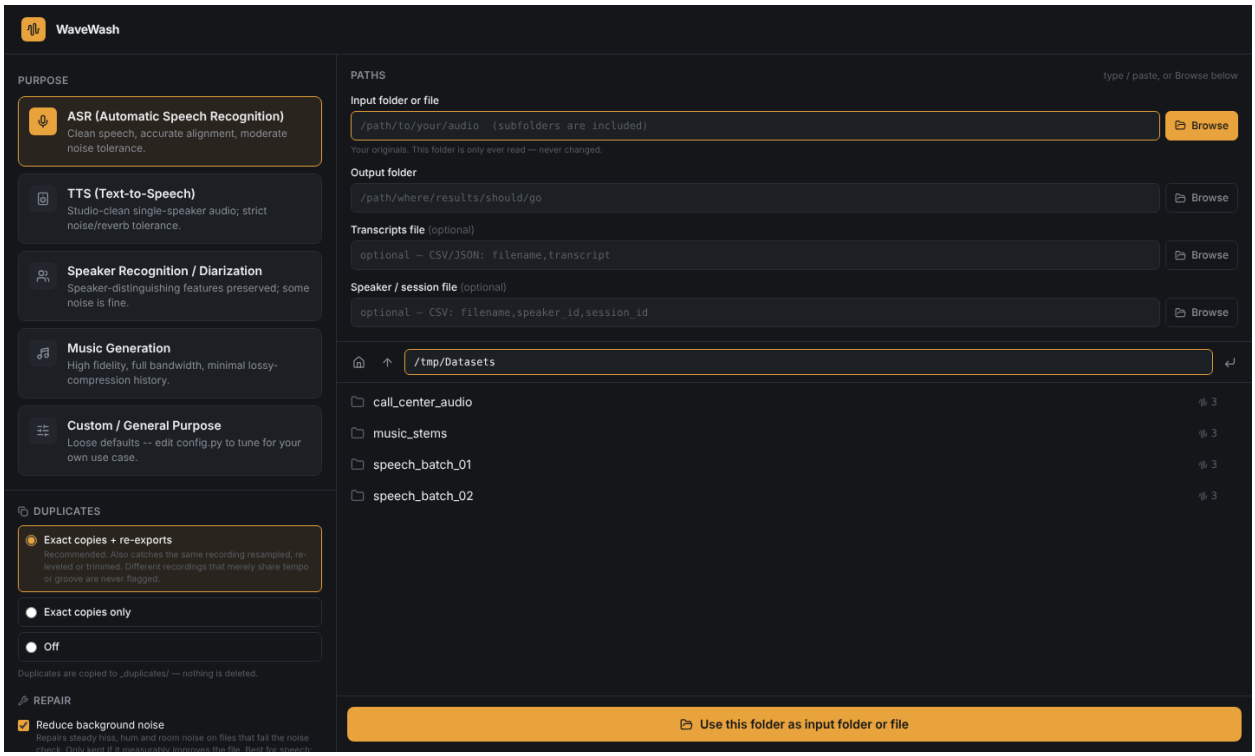
2. Start the app

```
wavewash app
```

Your browser opens by itself. If not, open `http://127.0.0.1:8420`. Keep the Terminal window open while you work. To stop WaveWash, press Ctrl+C in that window.
WaveWash runs only on your own computer. Nothing is uploaded anywhere and there is no login.

3. Use the screen, step by step

Before you start: put your audio files (.wav) into one folder. Make a second, empty folder for the results.



The setup screen. Purpose on the left, folders on the right.

Step 1. Pick your purpose (top left)

Choice	Use it for
ASR	Training speech-to-text. Clean speech, moderate noise is fine.
TTS	Making a voice. Needs studio-clean, one speaker. Strictest on noise.
Speaker Recognition	Telling people apart. Some noise is fine.
Music Generation	Music. Needs full quality, no heavy compression.
Custom / General	Anything else. Choose this if you are unsure.

The purpose sets how strict the checks are.

Step 2. Tell it where the files are (top right)

• **Input folder or file**: click **Browse** and click through folders until you reach your audio. Subfolders are included.
• **Output folder**: click **Browse** and choose your empty results folder. It must be a different folder from the input. System folders such as /etc are refused.
• **Transcripts file** and **Speaker / session file**: optional. Skip them unless you have a CSV with text or speaker names.

Step 3. Duplicates (left side)

• **Exact copies + re-exports** (recommended): also catches the same recording that was resampled, re-leveled or trimmed.
• **Exact copies only**: only identical files.
• **Off**: no duplicate check.
• Duplicates are copied aside. Nothing is deleted.

Step 4. Repair (left side)

• **Reduce background noise**: off by default. Turn on only for speech with steady hiss or hum. Leave off for music. The fix is kept only if the file measurably improves.
• **Also accept mp3 / flac / m4a / ogg**: off by default. Turn on if your files are not .wav.

Step 5. Run

Leave Quality Settings on **Simple** and press the **Run quality check** button at the bottom. Wait for the progress bar. The results page then appears.
Tip: try a small folder of about 20 files first.

4. Read your results

Open the output folder you chose. You will see:

Folder	What is inside
clean	Files that passed, or were repaired and then passed the check again.
_error	Rejected files, in sub-folders named by the reason (for example low_snr, clipping, corrupt_or_unreadable).
_duplicates	Duplicates that were found. These are copies, nothing is deleted.
_logs	The full report: audit_log_*.csv and .json (one row per file) and summary_*.md.

Open the `.csv` in Excel. Each row shows what was found in a file, what was fixed, and why it was rejected.
Important: "Repaired" means the file passed the check again. It does not mean the audio is restored to perfect. Listen to a few files in `_error` to make sure the tool is rejecting the right ones.

5. Using the command line instead

No browser needed. Example for speech-to-text:

```
wavewash run --input ./raw_audio --output ./out --purpose asr
```

Other commands: `wavewash doctor` (health check) and `wavewash --version`. Add `--help` to any command to see all options.

6. Common problems

Problem	What to do
Page does not open	Open http://127.0.0.1:8420 yourself. Make sure the Terminal window running wavewash app is still open.
"Output folder can't be inside the input folder"	Choose a separate, empty output folder.
"Choose a normal data folder"	You picked a system folder. Pick one in your Documents or Desktop.
No files found	Files must be .wav, or turn on the mp3/flac/m4a/ogg box.
Good files rejected	Pick a looser purpose (Custom) or use Advanced settings. Please report it on GitHub with the row from the csv.
Port already in use	Close the older WaveWash window, then start again.

7. What WaveWash can and cannot do

It checks and fixes:

• Background noise level, clipping (loud peaks cut off), DC offset, loudness
• Files that were secretly compressed (mp3-like) and saved as wav
• Broken, empty or unreadable files
• Exact, resampled and near-duplicate files
• Speakers mixed in one file (a rough guess, not exact)
• Reverb (shown as a warning only, because the estimate is not reliable)

Please be aware:

• This is an **alpha**. It uses classic signal processing, not machine learning, and it is not new research.
• It has been tested on synthetic audio and on Windows, macOS and Linux with automatic tests. It has **not yet** been measured on large real-speech datasets.
• Whisper, Silero voice detection and language detection are optional and not yet tested.
• Speed and memory on very large datasets are not yet measured.
• Whether cleaned data makes a trained model better is not yet measured.

Report a problem: https://github.com/BoobalanArjunan/wavewash/issues. A small example file and the csv row for it help the most.